

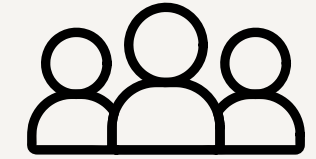


CodingLab

SWE-4404

SPL-II • Group-10

Team Members



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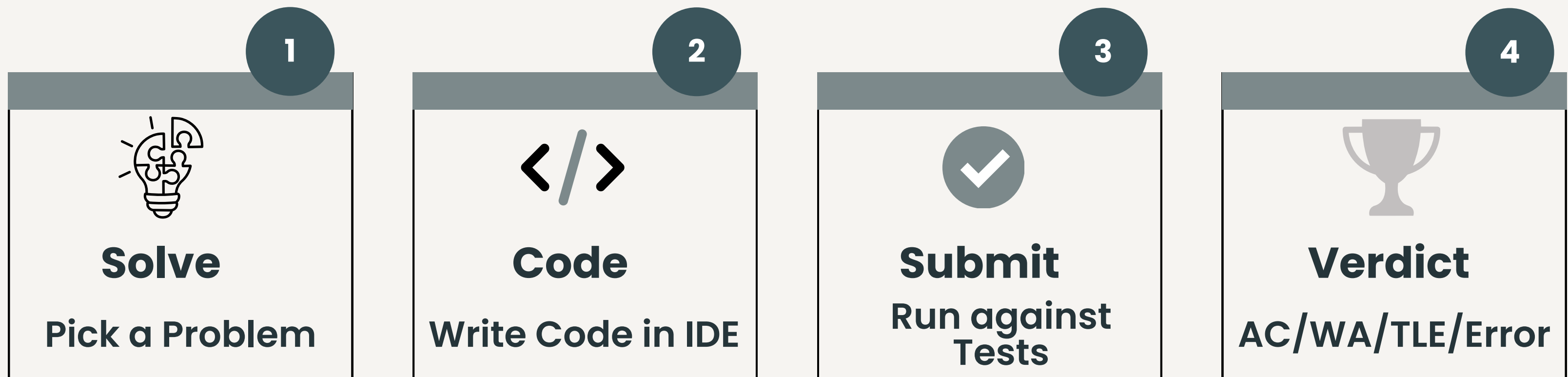
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230042144

Project Overview

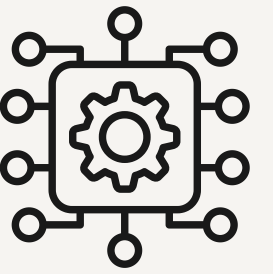
CodingLab

A web-based coding practice platform built for university students with role-based access, automated judging and full progress tracking

How it works



Three-Role Architecture



Student

Browse & Solve

Submit Code

Leaderboard

Progress



Problem Setter

Create Problems

Set Test Cases

Submit for
Review

Analytics



Admin

Approve / Reject

Manage Users

Platform
Stats

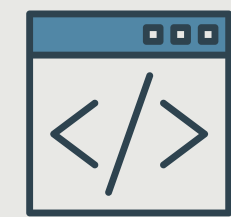
Contests

Student Features



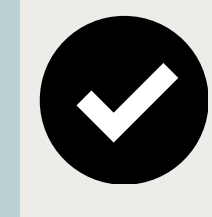
Discover Problems

Browse problems; filter by difficulty or search by title and topic tag



In-Browser Code Editor

Write and submit codes directly with syntax highlighting



Instant Judge Verdict

Judge0 evaluation with verdicts, time & memory stats



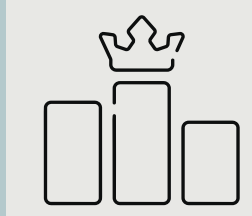
Progress Dashboard

View total solved, success rate, difficulty stats, and progress charts



Bookmarks & History

Bookmark problems for later and review submission history



Leaderboard

Compare rank against classmates - motivating healthy academic competition

Problem Setter Features



Create & Manage Problems

Define title, description, difficulty, tags, and test cases (sample + hidden); set time & memory limits & video solution



Submit Problems for Review & Approval

Submit completed problems for admin review, track status (Draft / Pending / Approved / Rejected), and refine with feedback for resubmission



Track Submissions

See how many students solved their approved problems. View submission statistics: acceptance rate, most common errors.



Problem Setter Dashboard

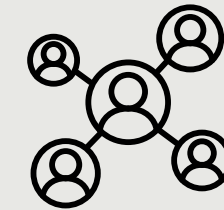
View total problems created & submissions; quickly edit pending/rejected problems

Admin Features



Moderate Problems

Review all pending problems submitted by problem setters. Edit or delete any inappropriate problem.



Manage Users

View all users with roles, manage role changes (Student ↔ Problem Setter), and approve or reject problem setter requests.



Create & Manage Contests

Create contests with title, duration, and select problems.
Edit or delete existing contests.



Platform Dashboard

Total users, problems (approved & pending), and all-time submissions, along with a recent activity feed showing new users, new problems, and new submissions.

System Architecture

CODE EXECUTION FLOW

**Student
Submits Code**



**Backend Sends
to Judg0 API**



**Judge0
Compiles and
Runs Safely**



**Verdict
Returned to
Platform**



**Result Shown
to Student**

PROBLEM APPROVAL FLOW

**Setter Creates
Problem**



**Submits for
approval**



**Admin
Reviews**



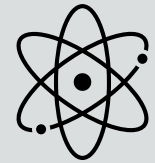
**Approved- >
Visible to all**



**Rejected- >
Feedback Sent**

Tools and Technologies

Frontend



React.js

Component-based UI,
fast rendering



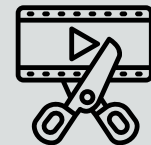
HTML/CSS/JS

Semantic markup &
responsive design



Tailwind CSS

Rapid responsive
styling



Monaco Editor

VS Code engine for
code workspace

Backend



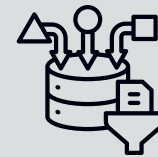
Node.js

JavaScript runtime for
server logic



express.js

Lightweight RESTful
API framework



Inngest

Submission queue
pipeline



Better Auth

Secure session &
RBAC auth

Database



MongoDB

NoSQL document store
for flexible schemas



Mongoose

ODM for schema
validation & queries

Service and Tools

JUDGE0

Judge0 API

Sandboxed compilation &
execution



Git/Github

Version control &
source submission



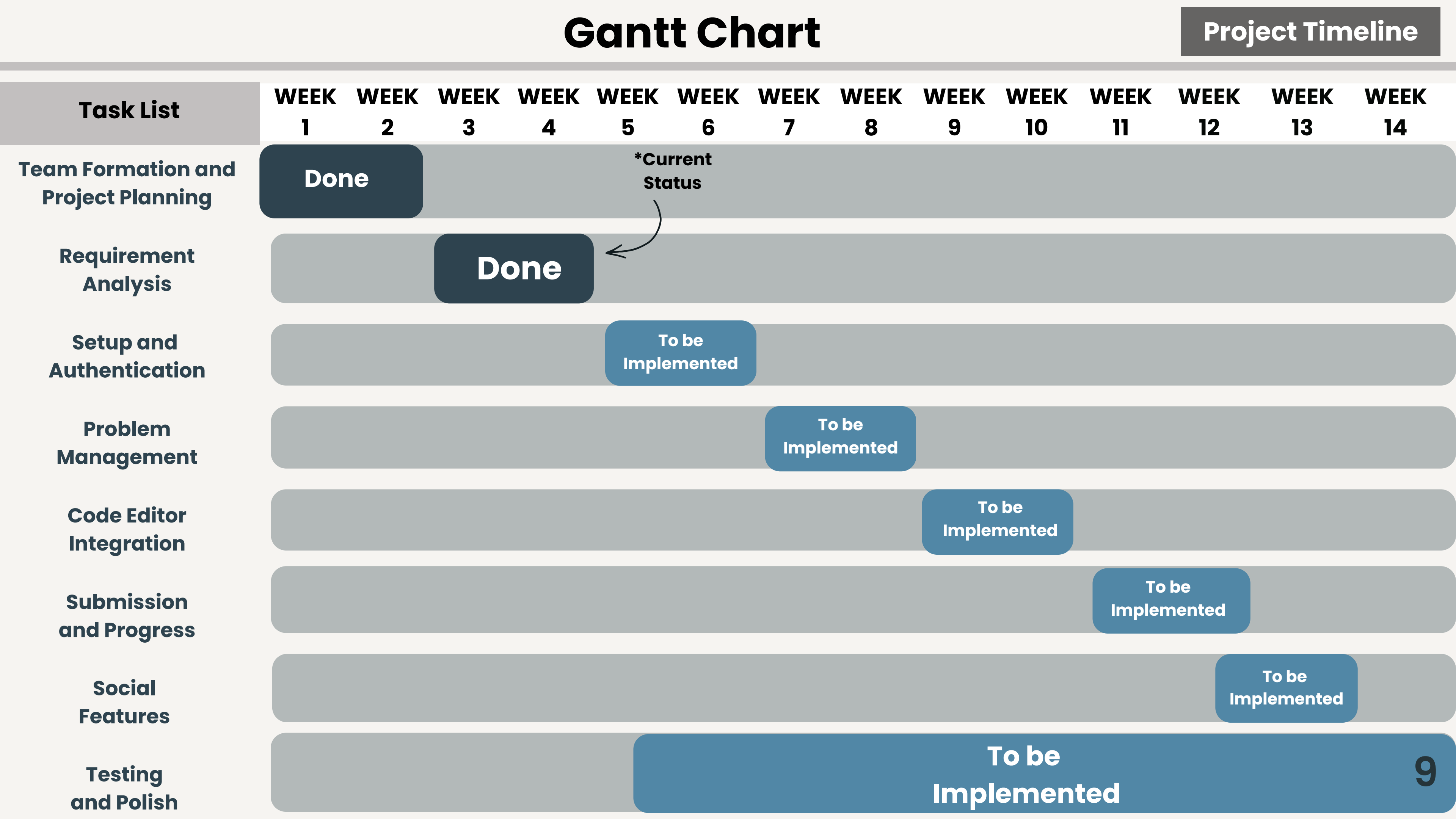
Cloudinary

Media storage & video
streaming



Docker

Isolated Judge0
hosting environment



Thank You